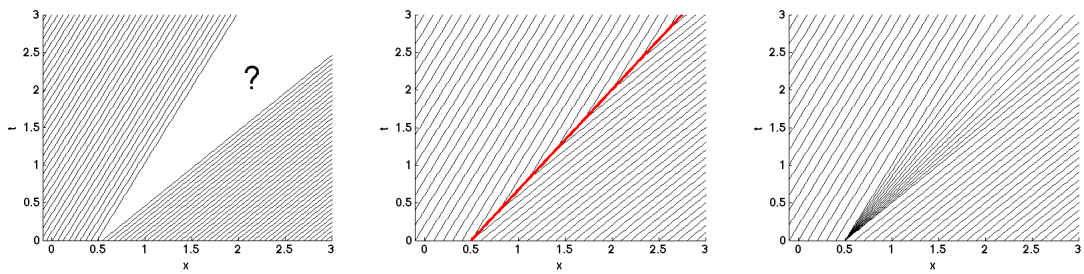

(a) Diverging characteristics (b) A "shock" instead of diverging characteristics (c) A smooth solution

Figure 8.6: The Burgers Equation: Characteristics for the Riemann problem of Example 8.4 with $x_0 = 0.5$.

the discontinuous solution (8.25) with $s = \frac{1}{2}(u^l + u^r)$ is a valid weak solution! Figure 8.6b shows this solution for the case $u^l < u^r$. On the other hand it may be seen that for $u^l < u^r$ a *continuous* solution to (8.23), (8.24) can be constructed (convince yourself that $u = x/t$ satisfies the Burgers equation!):

$$u(x,t) = \begin{cases} u^l \ , & x - x_0 < u^l t \ , \\ (x - x_0)/t \ , & u^l t \le x - x_0 \le u^r t \ , \\ u^r \ , & x - x_0 > u^r t \ . \end{cases}$$

Figure 8.6c shows a plot of this solution. So clearly, in this case, we can have more than one solution!

Example 8.4 clearly demonstrates that the class of weak solutions is too large. We have proved by counterexample that weak solutions are not in general unique. (Recall that a well-posed problem needs to have a unique solution.) Both solutions, which are illustrated in fig. 8.6 in terms of the corresponding characteristics are perfectly acceptable weak solutions to the same problem! Obviously then, weak solutions are not unique. We need additional conditions to identify meaningful solutions.

8.4 Entropy Solutions

As we have seen, the class of weak solutions is too large: Weak solutions are not in general unique. We must therefore, if possible, refine our notion of solution to a point where we can guarantee a unique solution, which is also consistent with whatever reasonable physical principles we may consult along the way. It turns out that for scalar conservation laws, ones can produce a watertight theory. For systems of equations, such as the Euler equations of section 8.1, there is, except for certain special cases, no complete theory of well-posedness.

One physical principle that we may use to narrow down the admissible class of weak solutions to conservation laws, is that of *viscosity*. In fact, many conservation laws, such as the Euler equations, are idealizations in that they model physical systems which do contain (perhaps very small) viscous effects which have been neglected. It seems therefore reasonable to seek such solutions to these problems that can be obtained as the zero-viscosity limit of a viscous perturbation. This is the concept of *vanishing viscosity*.

Recall from our previous discussion (e.g. heat equation) that visous effects can often be modeled by second-derivative terms. (Heat conduction is such an example.) For example, a

viscous perturbation of a scalar, one-dimensional conservation law can be written

$$u_t^\varepsilon + (f(u^\varepsilon))_x = \varepsilon u_{xx}^\varepsilon, \qquad \varepsilon > 0. \tag{8.26}$$

Loosely speaking, the aim is to define the solution to the unperturbed problem, i.e.

$$u_t + (f(u))_x = 0, \tag{8.27}$$

as the limit $u^\varepsilon \to u$, as $\varepsilon \to 0$. It is known that the viscous problem (8.26) admits a unique solution[5], and we have already pointed out that this limit solution would appear to make sense from a physical perspective. A further corroboration is obtained from thermodynamics: We can connect the vanishing viscosity solution to an entropy principle, which, for many cases describing physical systems, is consistent with entropy inequalities observed in nature (second law of thermodynamics!). The Euler equations of compressible inviscid flow are again a good example. It is known that entropy is a conserved quantity along streamlines in smooth regions, but increases across shock discontinuities.

We briefly outline the development of a theory of entropy solutions for scalar one-dimensional equations and highlight the connection to the vanishing viscosity solution. Let $\eta : \mathbb{R} \to \mathbb{R}$ be any *convex* function. Multiply the conservation law (8.27) with η' to obtain

$$\eta'(u)u_t + \eta'(u)(f(u))_x = 0.$$

We see immediately that upon definig a new flux function, termed the *entropy flux* $F(u)$, such that $F' = \eta' f'$, any *smooth* solution of the conservation law (8.27) will also satisfy a conservation law for η:

$$\eta(u)_t + (F(u))_x = 0. \tag{8.28}$$

Again, such manipulations work only if the solution is smooth. What kind of entropy principle should we have for a weak solution, which may not even be continuous? First, we integrate (8.28) over a rectangle bounded by x_0, x_1 and t_0, t_1, to give

$$\int_{x_0}^{x_1} \eta(u(x,t_1))dx - \int_{x_0}^{x_1} \eta(u(x,t_0))dx + \int_{t_0}^{t_1} F(x_1,t)dt - \int_{t_0}^{t_1} F(x_0,t)dt = 0.$$

We prefer this form in the following, as this equality is meaningful even for weak solutions. (Note that no differentiability is required.)

A proper entropy function (and corresponding entropy flux) is required to have the property that the entropy behaves monotonically with time[6], i.e.

$$\int_{x_0}^{x_1} \eta(u(x,t_1))dx - \int_{x_0}^{x_1} \eta(u(x,t_0))dx + \int_{t_0}^{t_1} F(x_1,t)dt - \int_{t_0}^{t_1} F(x_0,t)dt \le 0 \tag{8.29}$$

for *any* convex entropy function η, and corresponding entropy flux F with $F' = f'\eta'$. We call such a solution, an *entropy solution*. While this appears a bit like an ad hoc requirement, with the physical entropy principle serving only as a loose motivation, we shall now show that, in fact, the vanishing viscosity solution is an entropy solution. We start from (8.26), which we multiply by η' to obtain

$$\eta'(u^\varepsilon)u_t^\varepsilon + \eta'(u^\varepsilon)(f(u^\varepsilon))_x = \varepsilon\eta'(u^\varepsilon)u_{xx}^\varepsilon. \tag{8.30}$$

Using the product rule of differentiation, this can be rewritten

$$\eta'(u^\varepsilon)_t + (F(u^\varepsilon))_x = \varepsilon(\eta'(u^\varepsilon)u_x^\varepsilon)_x - \varepsilon\eta''(u^\varepsilon)(u_x^\varepsilon)^2.$$

[5]For exmaple, as a suitably formulated initial value problem. We will not discuss in detail the precise setting.

[6]It is worth pointing out that the combination of a convex function and the in equality with $\le$ in eq. (8.29) is somewhat arbitrary. We could replace this with "concave" and $\ge$. The important property is monotonicity. Note that while for physical entropy this latter definition is used more often (after all, entropy *increases* over time), the mathematical theory of entropy solutions is usually defined with a convex entropy. One can easily obtain the "mathematical" entropy from the "physical" entropy s by changing the sign: $s \mapsto -s$.

Again, performing the same integration as before, and using $(\eta' u_x^\varepsilon)_x = \eta(u^\varepsilon)_{xx}$, we have

$$\int_{x_0}^{x_1} \eta(u^\varepsilon(x,t_1))dx - \int_{x_0}^{x_1} \eta(u^\varepsilon(x,t_0))dx + \int_{t_0}^{t_1} F(x_1,t)dt - \int_{t_0}^{t_1} F(x_0,t)dt =$$
$$\varepsilon \left\{ \int_{t_0}^{t_1} \eta(u^\varepsilon(x_1,t))_x dx - \int_{t_0}^{t_1} \eta(u^\varepsilon(x_0,t))_x dx - \int_{x_0}^{x_1} \int_{t_0}^{t_1} \eta''(u^\varepsilon)(u_x^\varepsilon)^2 dxdt \right\}.$$

For bounded initial data, it may be shown that the initial value problem for perturbed problem has a uniformly bounded solution (in ε), and that the first term on the right-hand side vanishes as $\varepsilon \to 0$. Furthermore, the second term will be non-positive for a convex entropy η. Hence the inequality (8.29) holds in the limit $\varepsilon \to 0$.

We merely state the general definition of entropy solution, which is motivated by these thoughts. For general systems we first note that a convex function $\eta : \mathcal{D} \to \mathbb{R}$ is called an entropy of the system (8.10) if there exist d functions $F_i : \mathcal{D} \to \mathbb{R}$, $i = 1, \ldots, d$, such that

$$\nabla^T \eta(\mathbf{u}) D_{\mathbf{u}} \mathbf{f}_i(\mathbf{u}) = \nabla^T F_i(\mathbf{u}) \ , \qquad i = 1, \ldots, d \ , \tag{8.31}$$

where all gradients are with respect to $\mathbf{u}$. ($D_{\mathbf{u}}\mathbf{f}_i(\mathbf{u})$ is the Jacobian matrix A_i.) This is the generalization of the entropy flux for the scalar one-dimensional case, which satisfied $F' = f'(u)\eta'(u)$. It is then straightforward to see that we satisfy an additional conservation law for the entropy, i.e.

$$\partial_t \eta(\mathbf{u}) + \sum_i \frac{\partial F_i(\mathbf{u})}{\partial x_i} = 0.$$

This is again very similar to the scalar case (cf. eq. (8.28)). Multiplying the corresponding viscous perturbation (i.e. the analog of (8.30)) with smooth non-negative test functions, integrating by parts, and using arguments similar to those made for the simple example above, we can motivate the following definition of the *entropy solution*[7]:

Definition 8.4 *A weak solution* $\mathbf{u}$ *to the initial value problem* (8.19)–(8.20) *is called an* entropy solution*, if, for all entropy functions* $\eta(\mathbf{u})$ *and corresponding entropy fluxes* $F(\mathbf{u})$*, which satsify* (8.31)*, and for all* $v \in C_0^1(\mathbb{R}^d \times [0,\infty))$ *,* $v \geq 0$*,* $\mathbf{u}$ *satisfies*

$$\int_0^\infty \int_{\mathbb{R}^d} \left(\eta(\mathbf{u}) v_t + \sum_{i=1}^d F_i(\mathbf{u}) v_{x_i} \right) dxdt + \int_{\mathbb{R}^d} v(x,0)\eta(u_0(x))\, dx \geq 0 \ .$$

Note that, for systems, finding an entropy function is in general not trivial, as eq. (8.31) presents a system of $m \times d$ partial differential equations for the $(d+1)$ unknown functions η and F_i. In the scalar case, however, any convex function is an entropy, upon taking for the flux F_i any primitive of the function $\eta'(u)f_i'(u)$. (As we have noted before.)

In a derivation similar to that which lead to the Rankine-Hugoniot jump condition, one can show that a piecewise smooth weak solution $\mathbf{u}$ is an entropy solution if and only if it satsifies the jump inequality

$$n_t \left(U(\mathbf{u}_+) - U(\mathbf{u}_-) \right) + \sum_{i=1}^d n_{x_i} \left(F_i(\mathbf{u}_+) - F_i(\mathbf{u}_-) \right) \leq 0 \tag{8.32}$$

across discontinuities.

To summarize, a piecewise C^1 function $\mathbf{u}$ is a weak entropy solution of (8.10) if

[7] The relation between the strong form and the weak form here is similar to the relation between the strong solution and weak solution: One can get from one to the other by integration by parts.

- $\mathbf{u}$ is a classical solution that satisfies (8.10) pointwise in regions where $\mathbf{u}$ is smooth enough.
- $\mathbf{u}$ satisfies the Rankine-Hugoniot shock conditions (8.22),
- $\mathbf{u}$ satisfies the entropy jump conditions (8.32)

Note in particular that in smooth regions where we can use classical solutions, the entropy satisfies a conservation law in the pointwise sense, i.e. $\partial_t \eta + \sum_i \partial_{x_i} F_i = 0$.

Example 8.5 The function $U = -\rho s$ is an entropy for the Euler equations, where s is proportional to the thermodynamic entropy function, i.e. $s \propto \log(p/\rho^\gamma)$. It is known from inviscid fluid mechanics that the entropy is a conserved quantity in smooth regions (it is constant along streamlines), but increases across compression shocks.

Example 8.6 For the 1D Burgers equations one may choose any convex function, for examle $\eta(u) = u^2$. From the entropy flux conditions we thus find $F_i(u) = \frac{2}{3}u^3$. From the Rankine-Hugoniot shock conditions we already know that for the Burgers equation the shock speed $s = -n_t/n_x$ (where $n = (n_t, n_x)$ is the normal to the line of discontinuity) is given by $s = \frac{1}{2}(u^l + u^r)$. The entropy shock conditions then give

$$\begin{aligned} s(u_+^2 - u_-^2) - \frac{2}{3}\left(u_+^3 - u_-^3\right) &\geq 0 \ , \\ \Leftrightarrow \frac{1}{2}(u_+ + u_-)(u_+^2 - u_-^2) - \frac{2}{3}\left(u_+^3 - u_-^3\right) &\geq 0 \ , \\ \Leftrightarrow \frac{1}{6}\left(u_- - u_+\right)^3 &\geq 0 \ . \end{aligned}$$

This means that we allow shocks only for $u_- > u_+$, i.e. we allow only discontinuous solutions whenever characteristics intersect. Going back to fig. 8.6, we choose the smooth solution depicted in 8.6c, rather than the discontinuous solution shown in 8.6b, as the discontinuity is not permissible.

It may be shown that entropy solutions are unique in the scalar case. (For example, for a suitably formulated initial value problem.) For nonlinear systems this has not been proved in general. However, the complete mathematical framework of entropy solutions for multidimensional nonlinear systems is beyond the scope of this introductory text. We should recall, however, the connection between the vanishing-viscosity idea, and the mathematical notion of entropy.

8.5 Further Reading

The first few sections of Godlewski and Raviart [8] are a very readable introduction to some of the concepts discussed here. Also, LeVeque [9] has a good introduction on conservation laws: Chapter 2 has a derivation of conservation laws. Some of the concepts related to weak solutions and entropy solutions are given in Chapter 11.

A more complete exposition of the theory can be found in [7]. For example the well-posedness of the scalar initial value problem is proved there. However, this is very hard reading.